

Continuous Casting Machine (CCM) - Standard Operating Procedures

Emergency Mold Breakout Response

Step 1. Stop the strand withdrawal drive immediately upon receiving a breakout warning from the breakout prediction system or sighting molten steel dropping below the mold table.

Step 2. Close the tundish slide gate or drop the stopper rod instantly to cut off the flow of liquid steel into the mold.

Step 3. Cut off the ladle slide gate to isolate the main steel source from the tundish.

Step 4. Activate the emergency cooling water dump valves to flood the affected strand zone and freeze the remaining liquid shell safely.

Step 5. Evacuate the immediate casting pit area and keep all personnel clear of the strand path until thermal contraction ceases.

Step 6. Inspect the copper mold plates, frame assembly, and containment rolls for thermal distortion and mechanical damage before scheduling a rebuild.